

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.57_k1)

Metric

$$ds^2 = -\frac{1}{4} r^2 dt \otimes dt + dr \otimes dr - \frac{(a_1 r^2 e^{(2t)} + a_2 r^2 + r^2 e^t) e^{(-t)}}{2(\rho^2 - 1)} d\rho \otimes d\rho + \frac{1}{2} (a_1 r^2 \rho^2 e^{(2t)} + a_2 r^2 \rho^2 + r^2 \rho^2 e^t) e^{(-t)} d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \rho, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{4} \sqrt{2} r\right) dt + \left(\frac{1}{2} \sqrt{2}\right) dr$$

$$n = \left(\frac{1}{4} \sqrt{2} r\right) dt + \left(-\frac{1}{2} \sqrt{2}\right) dr$$

$$m = \left(\frac{\sqrt{a_1 e^{(2t)} + a_2 + e^t |r|} e^{(-\frac{1}{2} t)}}{2 \sqrt{-\rho^2 + 1}}\right) d\rho + \left(\frac{1}{2} i \sqrt{a_1 e^{(2t)} + a_2 + e^t \rho |r|} e^{(-\frac{1}{2} t)}\right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{a_1 e^{(2t)} + a_2 + e^t |r|} e^{(-\frac{1}{2} t)}}{2 \sqrt{-\rho^2 + 1}}\right) d\rho + \left(-\frac{1}{2} i \sqrt{a_1 e^{(2t)} + a_2 + e^t \rho |r|} e^{(-\frac{1}{2} t)}\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{r}\right) dt + \left(\frac{1}{2} \sqrt{2}\right) dr$$

$$n = \left(-\frac{\sqrt{2}}{r} \right) dt + \left(-\frac{1}{2} \sqrt{2} \right) dr$$

$$m = \left(-\frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\sqrt{-\rho+1}|r|} \right) d\rho + \left(\frac{ie^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\rho|r|} \right) d\phi$$

$$\bar{m} = \left(-\frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\sqrt{-\rho+1}|r|} \right) d\rho + \left(-\frac{ie^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\rho|r|} \right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{r} \partial_t + \frac{1}{2} \sqrt{2} \partial_r$$

$$\Delta = -\frac{\sqrt{2}}{r} \partial_t - \frac{1}{2} \sqrt{2} \partial_r$$

$$\delta = -\frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\sqrt{-\rho+1}|r|} \partial_\rho + \frac{ie^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\rho|r|} \partial_\phi$$

$$\bar{\delta} = -\frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\sqrt{-\rho+1}|r|} \partial_\rho - \frac{ie^{\left(\frac{1}{2}t\right)}}{\sqrt{a_1e^{(2t)}+a_2+e^t}\rho|r|} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}(2a_2+e^t)}{2(a_1e^{(2t)}+a_2+e^t)r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}(2a_1e^t + 1)e^t}{2(a_1e^{(2t)} + a_2 + e^t)r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{2\sqrt{a_1e^{(2t)} + a_2 + e^t}\rho\sqrt{-\rho+1}|r|}$$

$$\beta = -\frac{\sqrt{\rho+1}(\rho-1)e^{\left(\frac{1}{2}t\right)}}{2\sqrt{a_1e^{(2t)} + a_2 + e^t}\rho\sqrt{-\rho+1}|r|}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}}{4r}$$

$$\gamma = \frac{\sqrt{2}}{4r}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{(4a_1a_2 - 1)e^{(2t)}}{2(a_1e^{(2t)} + a_2 + e^t)^2r^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{(4a_1a_2 - 1)e^{(2t)}}{4(a_1e^{(2t)} + a_2 + e^t)^2r^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{(4a_1a_2 - 1)e^{(2t)}}{2(a_1e^{(2t)} + a_2 + e^t)^2r^2}$$

$$\Lambda = \frac{(4a_1a_2 - 1)e^{(2t)}}{12(a_1e^{(2t)} + a_2 + e^t)^2r^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{(4a_1a_2 - 1)e^{(2t)}}{3(a_1e^{(2t)} + a_2 + e^t)^2r^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"